End-to-End Sales Data Warehouse Documentation

1. Executive Summary

This project focuses on building a modern Data Warehouse (DWH) to integrate sales data from two different sources: a CRM System and an ERP System.

The goal is to transform raw, inconsistent data into a high-quality Star Schema optimized for Business Intelligence (BI) reporting and analytics.

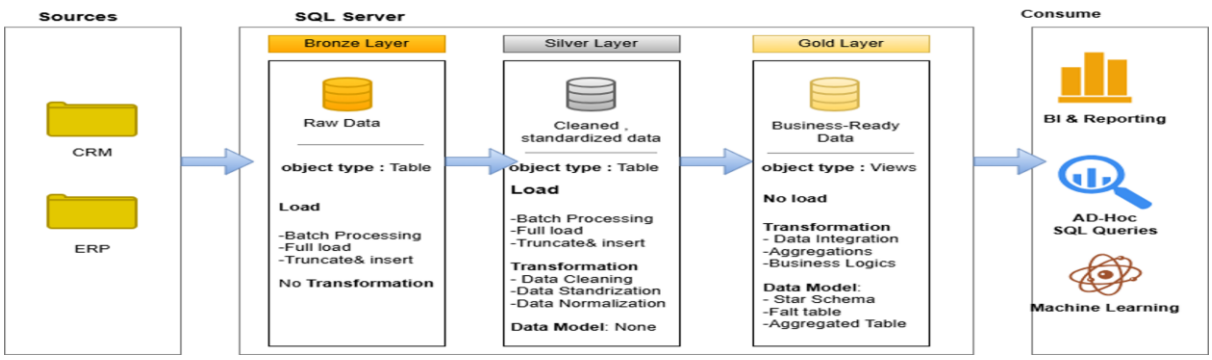
Key Objectives:

- Consolidate data from multiple CSV sources.
- Clean and standardize inconsistent data types and values.
- Model data into Dimensions and Facts (Star Schema).
- Ensure data quality through rigorous validation checks.

2. Data Architecture

The project follows the Medallion Architecture consisting of three layers:

Layer	Description	Object Type	Load Strategy
Bronze (Raw)	Stores data exactly as it comes from the source systems	Table	Batch Processing / Full Load / Truncate & Insert
Silver (Cleansed)	Data is cleaned, deduplicated, and normalized	Table	Batch Processing / Full Load / Truncate & Insert
Gold (Curated)	Data is modeled into a Star Schema for reporting	Views	No Load — Views refresh on query



3. Implementation Details

3.1 Bronze Layer — Ingestion

Item	Details
Source Data	CSV files: cust_info, prd_info, sales_details, CUST_AZ12, LOC_A101, PX_CAT_G1V2
Technique	BULK INSERT within Stored Procedures — for rapid, efficient data loading
Outcome	Raw tables created with original schemas, no transformations applied

3.2 Silver Layer — Transformation & Cleansing

In this layer, significant data engineering was performed using T-SQL

Transformation	Details
Standardization — Gender	Unified values: 'F', 'FEMALE' → 'Female'
Standardization — Marital Status	Converted: 'S', 'M' → 'Single', 'Married'
Standardization — Country Codes	Mapped: 'DE' → 'Germany', 'US' → 'United States'
Date Handling — Format	Converted integer dates (e.g. 20230101) to proper DATE format
Date Handling — Invalid Dates	Handled invalid / future birthdates by setting them to NULL
Deduplication	Applied ROW_NUMBER() to remove duplicate customer records based on most recent create_date
Logic Correction	Recalculated Sales Amount where values were missing or negative using: Quantity × Price

3.3 Gold Layer — Data Modeling

The data was restructured into a Star Schema:

Object	Type	Description
dim_customers	Dimension	Enriched customer data combined with ERP demographics (first/last name, gender, marital status, birthdate, country)
dim_products	Dimension	Merged product details with categories. Filtered for active products only (end_date IS NULL)
fact_sales	Fact Table	Contains transactional sales data linked to all dimensions via Surrogate Keys

4. Data Quality & Validation

Before finalizing the data, the following checks were executed:

#	Check	Description
1	Null Checks	Ensured no Primary Keys or Surrogate Keys are null across all dimension and fact tables
2	Referential Integrity	Validated that every sale in the Fact table links to a valid Customer and Product dimension record
3	Business Logic	Verified that Sales Amount is never negative across all transactions
4	Date Logic	Confirmed that Product End Date is not earlier than the Product Start Date

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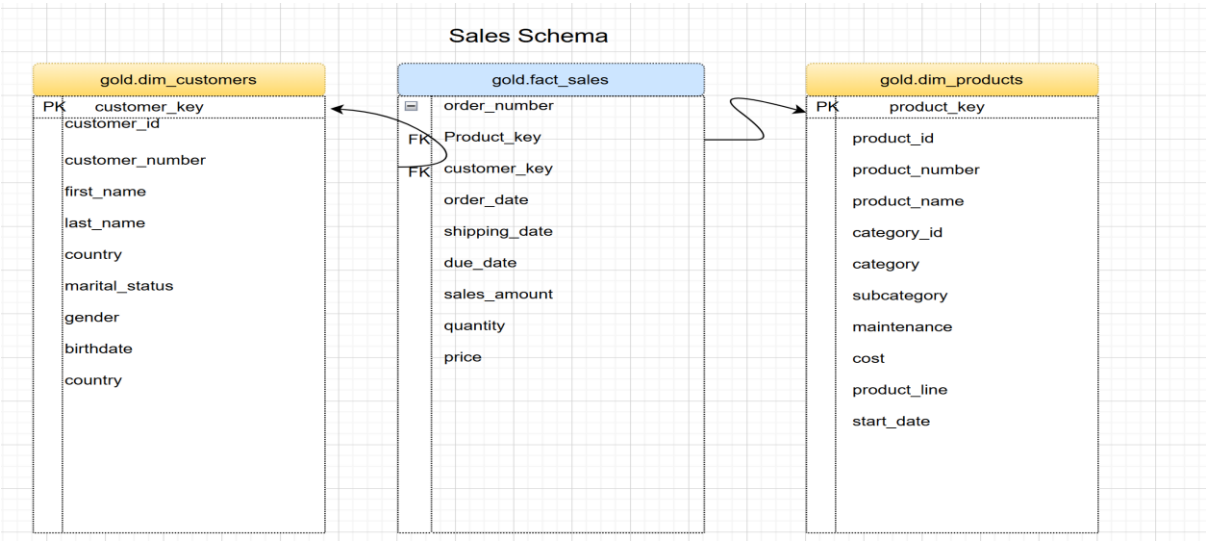
5. Entity Relationship Diagram (ERD)

The Star Schema is centered on the fact_sales table, with dimension tables connected via Surrogate Keys.

Object	Role	Connection
fact_sales	Fact Table (Center)	Contains FK references to all dimension tables
dim_customers	Dimension	Linked via customer_key → fact_sales.customer_key
dim_products	Dimension	Linked via product_key → fact_sales.product_key

Schema Diagram:



6. Conclusion

The Data Warehouse is now fully operational. The data is clean, consistent, and structured — ready for visualization tools like Power BI or Tableau to build dashboards and deliver actionable business insights.